

Track A - Progress Report

Service Layer v1 + AI router, verified in QA

2026-09-28 - branch feature/service-layer

Locked inputs: canonical tenant, date rules 1-6, QA simulation

1. Integration first

Rick's e7ef7cf (CR reconciliation + Outstanding) and de769cc (APR clear/batch/void) were merged BEFORE any Track A code, so Q3 builds on his Outstanding structures instead of duplicating them. Conflict calls: kept zip-to-tz (Rick) + authMid (ours); kept HOA-scoped writes + DUP retry; discarded Alex's temporary PRINT-CHECKS hack; APR list combines tenant filter with Status IN (POSTED,VOID).

2. Fase 0 - AI safety (tenant leak closed)

- POST /api/ai-filter had no HOA scope: every user saw all HOAs (72 residents instead of 11 for HOA5).
- Now: requireHoaScope, parameterized HOALicenseNumber filter, fail-closed answer validation, tenant/scoped envelope, prompt+license logging.
- Admin X-HOA-ID=all keeps an explicit global view (scoped:false).

Tenant isolation proof (QA)

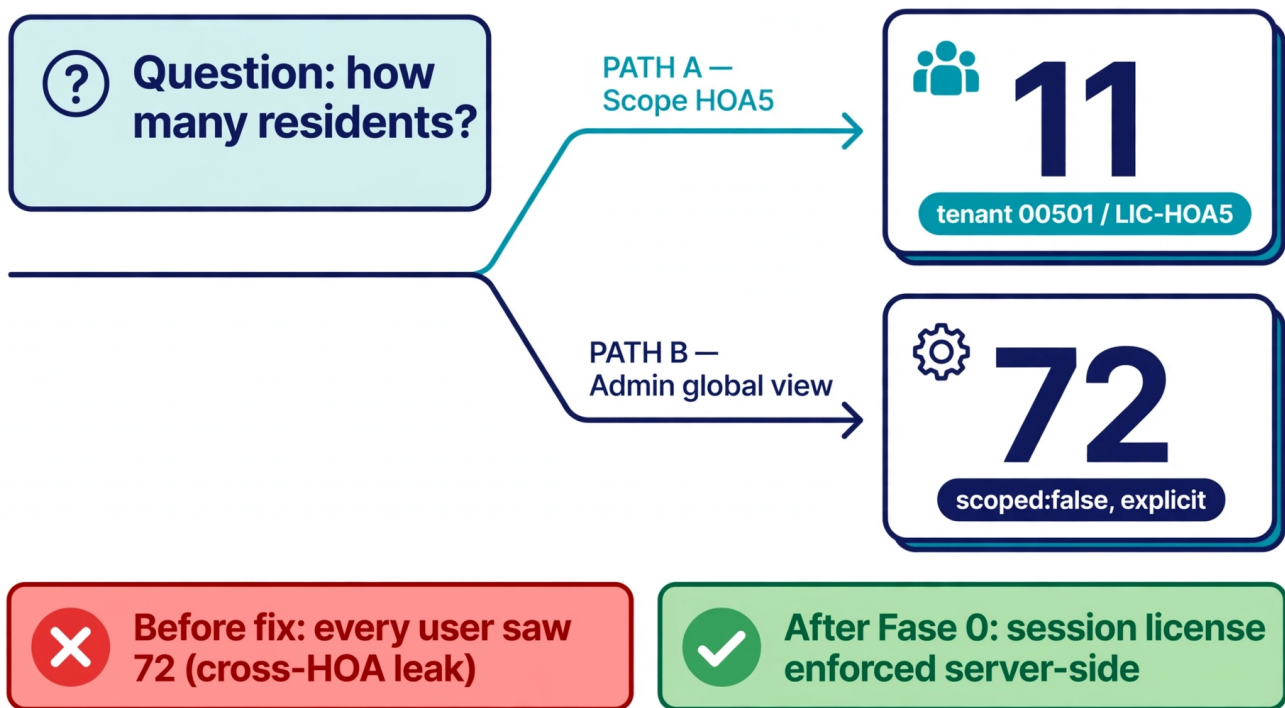
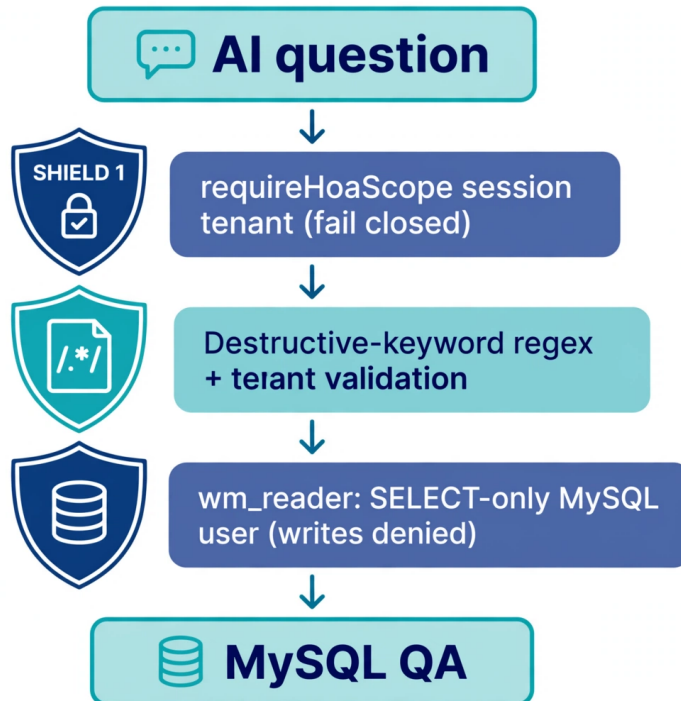


Fig. 1 - Tenant isolation proof: HOA5 sees 11, global sees 72.

Double shield: session tenant enforcement + destructive-keyword regex + wm_reader, a SELECT-only MySQL user (writes denied at privilege level, verified).

Double shield: read-only AI



Even a malicious prompt can only call a closed function with validated params.

Fig. 2 - Even a malicious prompt can only call a closed function with validated params.

3. Simulated tenant in QA (prod untouched)

- `hoa.client_id` populated: RL/GL/Ren = 00125, self-managed singles, DEV separate.
- Finding: `UNIQUE(client_id)` contradicted the approved model (one ClientID to many HOAs) - replaced by a plain index in QA ONLY. Prod keeps `UNIQUE` until Rick authorizes the change at real backfill.
- Settings Client ID# shows the canonical value; both identity fields read-only for licensees.

4. Date standard implemented (rules 1-6)

- `resolveFiscalWindow()`: explicit range > `FiscalYearSetup` (per HOA) > `hoa.fiscal_year_start` > `calendar` (flagged). Plus `containing(date)` mode for clear/pay flows.
- Per-ZIP timezone cache (the old global cache pinned the first HOA's zone on all others).
- Migrated 9 spots: ledger, cash-flow page, check/deposit clear, 3x APR, escrow, YTD, monthly-GL.
- QA FY2026 = `calendar`, so all windows are identical - verified with real data (escrow GL 41700 = known fixture).

5. Registry v1 - 7 functions at `/api/svc/*`

- READY: account-history Q1+Q2 (balance 300 + 1 payment), outstanding-checks Q3 (2 checks, 312.50), gl-transactions Q5 (4 txns, net 12500), vendor-invoices Q7.
- STUB (honest, no invented numbers): period-diff Q4, anomalies Q8.
- DISABLED: violations Q6 returns 403, waits for FL definitions.
- All reads via `readOnlyPool`; concrete HOA required; envelope carries tenant + resolved params + lineage.

Service registry v1: 7 functions

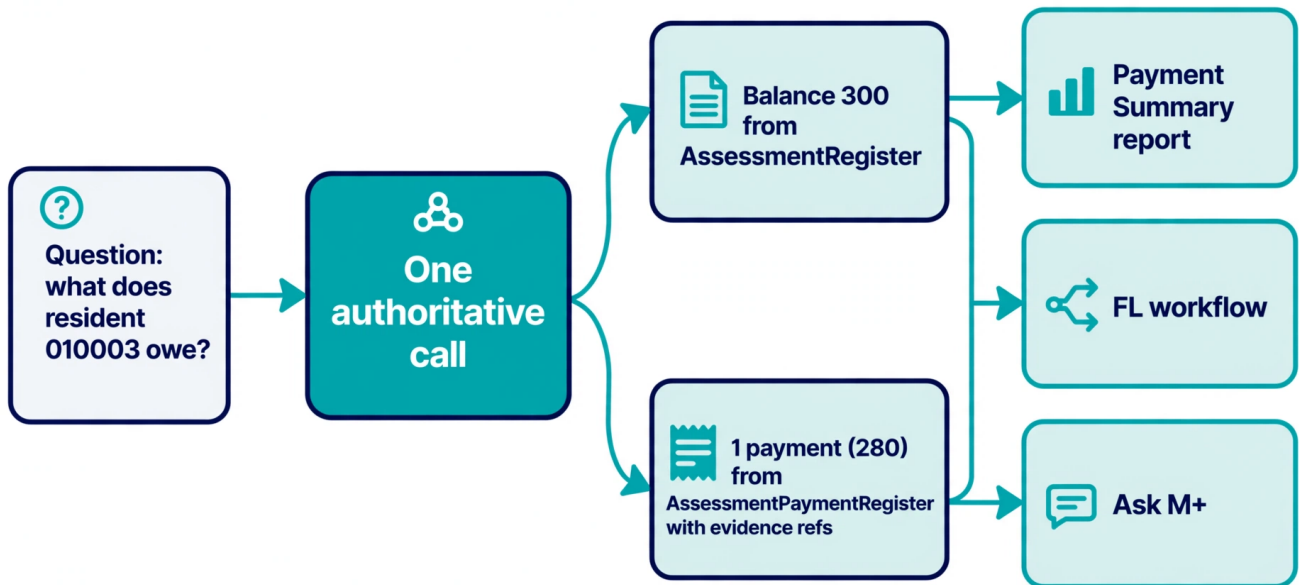
 `/api/svc**`

						
account-history	outstanding-checks	period-diff	gl-transactions	vendor-invoices	violations	anomalies
Q1+Q2	Q3	Q4	Q5	Q7	Q6	Q8
READY	READY	STUB	READY	READY	DISABLED waits FL	STUB

 Standard envelope: tenant + params + summary + lineage

Fig. 3 - Frozen contract: 7 entries, 4 ready, 2 stubs, 1 disabled.

getResidentAccountHistory: one function, Q1+Q2



- Same number, three consumers.

Fig. 4 - Q1+Q2 reference: same number, three consumers.

6. AI router - providers

- Level 1 local classifier: 11 intents, free, instant.
- Level 2 OpenRouter cheap LLM (~\$0.001/translation): prompt + catalog only, never rows, never SQL; Q6 banned even for the LLM. Driver by env (model = one variable).
- Level 3 legacy free-SQL path: DEPRECATED, removed after golden set.
- Golden set documented (8 questions with expected routes/values).

AI Router cascade

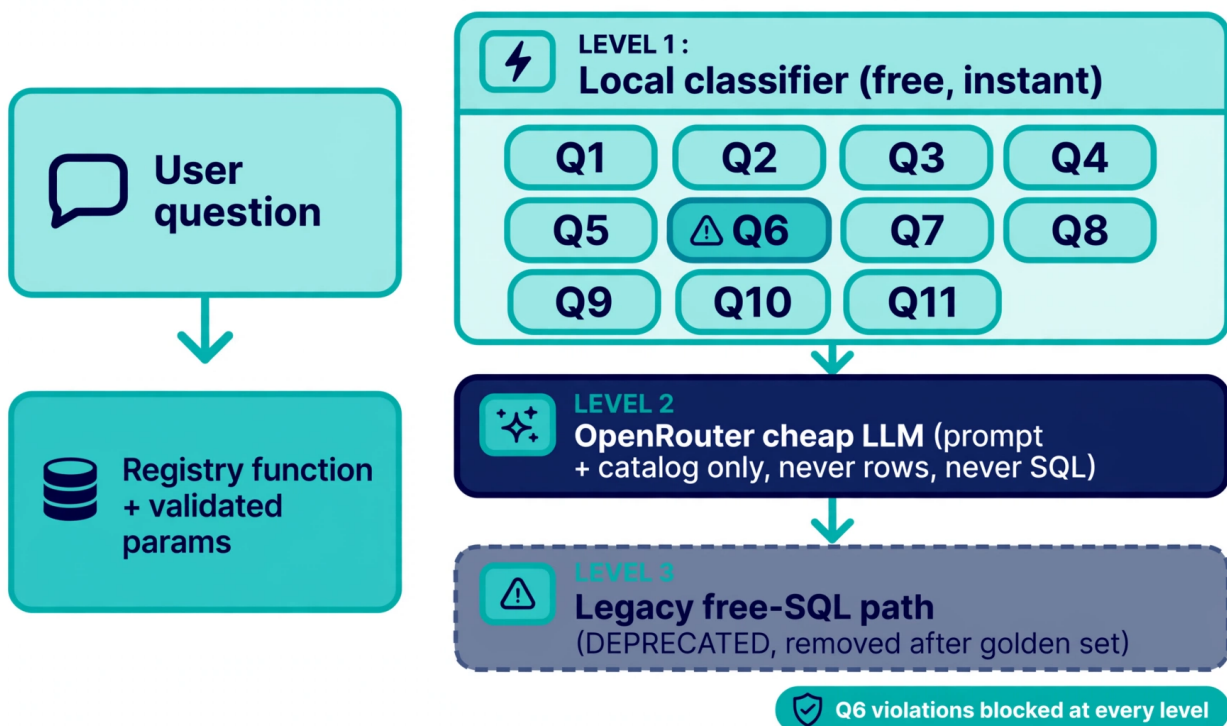


Fig. 5 - Cascade: local -> cheap LLM -> legacy (deprecated).

7. Commits and open items

Branch feature/service-layer: merge Rick, AI fix, auth UI, Fase 0, tenant, dates, registry, router. Open: real ClientID Excel (pre-prod) + drop UNIQUE in prod; 12-level catalog (pending Rick/Hal); FL definitions for Q6/overdue/histories; Ask M+ skeleton consumes the registry next.